U.PORTO
FEUP FACULDADE DE ENGENHARIA UNIVERSIDADE DO PORTO
DEI DEPARTAMENTO DE ENGENHARIA INFORMÁTICA

U.PORTO
FC FACULDADE DE CIÊNCIAS UNIVERSIDADE DO PORTO
[dcc] DEPARTAMENTO DE CIÊNCIA DE COMPUTADORES FACULDADE DE CIÊNCIAS DA UNIVERSIDADE DO PORTO

Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

4 - Semantics of First Order Logic - Solutions

4.1) Consider the language $\mathcal{L}$, a language of first order with equality and such that $\mathcal{F}_0 = \{a, b\}$, $\mathcal{F}_1 = \{g\}$, $\mathcal{F}_2 = \{f, h\}$, $\mathcal{R}_1 = \{R, S\}$ and $\mathcal{R}_2 = \{P, Q\}$). Let $\mathcal{A}$ be a structure of the language $\mathcal{L}$ defined by:

- the domain of $\mathcal{A}$ is the set of natural numbers $\mathbb{N} = \{0, 1, \ldots\}$
- $a^{\mathcal{A}} = 1$, $b^{\mathcal{A}} = 0$
- $g^{\mathcal{A}}(n) = n + 4$, $n \in \mathbb{N}$
- $h^{\mathcal{A}}(n, m) = n + m$, $f^{\mathcal{A}}(n, m) = n \times m$, $n, m \in \mathbb{N}$
- $S^{\mathcal{A}} = \{n \in \mathbb{N} \mid n \text{ is odd}\}$,$R^{\mathcal{A}} = \{n \in \mathbb{N} \mid n \text{ is even}\}$
- $P^{\mathcal{A}} = \{(n, m) \in \mathbb{N}^2 \mid n > m\}$
- $Q^{\mathcal{A}} = \emptyset$

If φ is each of the following propositions, determine **justifying** whether it is true or false in $\mathcal{A}$, i.e. if $\mathcal{A} \models \varphi$ or $\mathcal{A} \not\models \varphi$.

(a) $\forall x \exists y\ P(y, x)$

Solution: $\mathcal{A} \models \forall x \exists y\ P(y, x)$. The explanation is the following: $\mathcal{A} \models \forall x \exists y\ P(y, x)$, if for every variable interpretation s, $\mathcal{A} \models_s \forall x \exists y\ P(y, x)$. This happens if for every $a \in \mathbb{N}$ we have $\mathcal{A} \models_{s[x \mapsto a]} \exists y\ P(y, x)$, that is, if for every $a \in \mathbb{N}$ there exists $b \in \mathbb{N}$ such that $\mathcal{A} \models_{s[x \mapsto a][y \mapsto b]} P(y, x)$, that is if for every $a \in \mathbb{N}$ there exists $b \in \mathbb{N}$ such that $(s[x \mapsto a][y \mapsto b](y), s[x \mapsto a][y \mapsto b](x)) \in P^{\mathcal{A}}$. This is equivalent to say that for every $a \in \mathbb{N}$ there exists $b \in \mathbb{N}$ such that $b > a$. This is true for every a, just consider $b = a + 1$.

(b) $\exists x \forall y\ P(y, x)$

Solution: $\mathcal{A} \not\models \exists x \forall y\ P(y, x)$. The explanation is the following: $\mathcal{A} \models \exists x \forall y\ P(y, x)$, if for every variable interpretation s, $\mathcal{A} \models_s \exists x \forall y\ P(y, x)$. This happens if there exists $a \in \mathbb{N}$ such that $\mathcal{A} \models_{s[x \mapsto a]} \forall y\ P(y, x)$, that is, if there exists $a \in \mathbb{N}$ such that for every $b \in \mathbb{N}$ we have $\mathcal{A} \models_{s[x \mapsto a][y \mapsto b]} P(y, x)$, that is if there exists $a \in \mathbb{N}$ such that for every $b \in \mathbb{N}$ we have $(s[x \mapsto a][y \mapsto b](y), s[x \mapsto a][y \mapsto b](x)) \in P^{\mathcal{A}}$. This is equivalent to say that there exists $a \in \mathbb{N}$ such that for every $b \in \mathbb{N}$ we have $b > a$. And this is not true, since for every $a \in \mathbb{N}$ there is always a $b \in \mathbb{N}$ such that $b \leq a$, just take $b = 0$.

(c) $\forall x \exists y\ (f(x, y) = a)$

Solution: $\mathcal{A} \not\models \forall x \exists y\ (f(x, y) = a)$. The explanation is the following: $\mathcal{A} \models \forall x \exists y\ (f(x, y) = a)$ if for every variable interpretation s, $\mathcal{A} \models_s \forall x \exists y\ (f(x, y) = a)$. This happens if for every $b \in \mathbb{N}$

we have $\mathcal{A} \models_{s[x\mapsto b]} \exists y\ (f(x,y) = a)$, that is, if for every $b \in \mathbb{N}$ there exists $c \in \mathbb{N}$ such that $\mathcal{A} \models_{s[x\mapsto b][y\mapsto c]} (f(x,y) = a)$, which follows it if for every $b \in \mathbb{N}$ there exists $c \in \mathbb{N}$ such that $f^{\mathcal{A}}(s[x \mapsto b][y \mapsto c](x), s[x \mapsto c][y \mapsto c](y)) = s[x \mapsto b][y \mapsto c](a)$. This is equivalent to say that for every $b \in \mathbb{N}$ there exists $c \in \mathbb{N}$ such that $b \times c = 1$. This is not true in $\mathbb{N}$: just note that if we take $b = 2$ then for every $b \in \mathbb{N}$, $2 \times b \neq 1$.

(d) $\forall x \exists y\ P(y,x) \to \exists y P(y,y)$

Solution: $\mathcal{A} \not\models \forall x \exists y\ P(y,x) \to \exists y P(y,y)$. The explanation is the following: $\mathcal{A} \models \forall x \exists y\ P(y,x) \to \exists y P(y,y)$ if $\mathcal{A} \not\models \forall x \exists y\ P(y,x)$ or $\mathcal{A} \models \exists y\ P(y,y)$. The first option would be true if there was a variable interpretation s for which $\mathcal{A} \not\models_s \forall x \exists y\ P(y,x)$. This happens if for that variable interpretation, for some $a \in \mathbb{N}$ we have $\mathcal{A} \not\models_{s[x\mapsto a]} \exists y\ P(y,a)$. In turn, this happens if for any $b \in \mathbb{N}$ we have $\mathcal{A} \not\models_{s[x\mapsto a][y\mapsto b]} P(b,a)$. Now note that we can always choose $b = a + 1$, for any a, and for those values $\mathcal{A} \models_{s[x\mapsto a][y\mapsto b]} P(b,a)$, so $\mathcal{A} \models \forall x \exists y\ P(y,x)$. For the other option, if $\mathcal{A} \models \exists y\ P(y,y)$ then we would have that for some variable interpretation s and some value $c \in \mathbb{N}$, that $\mathcal{A} \models_{s[y\mapsto c]} P(c,c)$. However, there is no such value, since for any value c, $\mathcal{A} \not\models_{s[y\mapsto c]} P(c,c)$. Therefore $\mathcal{A} \not\models \exists y\ P(y,y)$.

(e) $\forall x\ S(x) \to \exists x\ S(x)$

Solution: $\mathcal{A} \models \forall x\ S(x) \to \exists x\ S(x)$. The explanation is the following: $\mathcal{A} \models \forall x\ S(x) \to \exists x\ S(x)$ if for every variable interpretation s, $\mathcal{A} \models_s \forall x\ S(x) \to \exists x\ S(x)$. This happens if $\mathcal{A} \not\models_s \forall x\ S(x)$ or $\mathcal{A} \models_s \exists x\ S(x)$. Now note that $\mathcal{A} \models_s \exists x\ S(x)$ if there exists $a \in \mathbb{N}$ such that $\mathcal{A} \models_{s[x\mapsto a]} S(x)$, that is, if there exists $a \in \mathbb{N}$ such that $s[x \mapsto a](x) \in S^{\mathcal{A}}$, which is equivalent to say that there exists $a \in \mathbb{N}$ such that a is odd. This is obviously true for any odd natural number.

(f) $\forall x\ (S(x) \vee R(x)) \to (\forall x\ S(x) \vee \forall x\ R(x))$.

Solution: $\mathcal{A} \not\models \forall x\ (S(x) \vee R(x)) \to (\forall x\ S(x) \vee \forall x\ R(x))$. the explanation is the following: $\mathcal{A} \models \forall x\ (S(x) \vee R(x)) \to (\forall x\ S(x) \vee \forall x\ R(x))$ if for every variable interpretation s, $\mathcal{A} \models_s \forall x\ (S(x) \vee R(x)) \to (\forall x\ S(x) \vee \forall x\ R(x))$. This happens if $\mathcal{A} \not\models_s \forall x\ (S(x) \vee R(x))$ or $\mathcal{A} \models_s \forall x\ S(x) \vee \forall x\ R(x)$. Now note that for every $a \in \mathbb{N}$, a is even or a is odd, which means that $\mathcal{A} \models_s \forall x\ (S(x) \vee R(x))$. Also, it is not true that every $a \in \mathbb{N}$ is odd, thus $\mathcal{A} \not\models_s \forall x\ S(x)$; and it is not true that every $a \in \mathbb{N}$ is even, thus $\mathcal{A} \not\models_s \forall x\ R(x)$. This implies that $\mathcal{A} \not\models_s \forall x\ S(x) \vee \forall x\ R(x)$.

(g) $\forall x(S(g(x)) \to S(x))$

Solution: $\mathcal{A} \models \forall x(S(g(x)) \to S(x))$. The explanation is the following: $\mathcal{A} \models \forall x(S(g(x)) \to S(x))$ if for every interpretation $\mathcal{A} \models_s \forall x(S(g(x)) \to S(x))$. This happens if for any $a \in \mathbb{N}$ we have $\mathcal{A} \models_{s[x\mapsto a]} S(g(a)) \to S(a)$. For this implication to be true if any number a plus 4 is odd, then so is a. This is true for any number.

4.2) Consider the language $\mathcal{L}$, a language of first order logic without equality and such that $F_0 = \{a\}$, $F_1 = \{f\}$, $F_2 = \{g\}$, $R_1 = \{R\}$ and $R_2 = \{S\}$. For each of the following formulas of the following formulas ϕ, create two structures $\mathcal{A}$ and $\mathcal{B}$ such that $\mathcal{A} \models \phi$ and $\mathcal{B} \not\models \phi$. These solutions are examples of structures, obviously several others woudl work too.

a) $\forall x(\exists y(S(x,y) \to R(f(y))))$

Solution: $\mathcal{A} = (\mathbb{N}, \cdot^{\mathcal{A}})$ where:
$a^{\mathcal{A}} = 1 \qquad f^{\mathcal{A}}(x) = x \times 0 \qquad g^{\mathcal{A}}(x,y) = x + y$
$R^{\mathcal{A}} = \{0\} \qquad S^{\mathcal{A}} = \emptyset$

$\mathcal{B} = (\mathbb{N}, \cdot^{\mathcal{B}})$ where:
$a^{\mathcal{B}} = 1 \qquad f^{\mathcal{B}}(x) = x \times 0 \qquad g^{\mathcal{B}}(x,y) = x + y$
$R^{\mathcal{B}} = \{x \mid x \text{ is prime}\} \qquad S^{\mathcal{B}} = \{(x,y) \mid x \leq y\}$

Note that $S(x,y)$ in $\mathcal{A}$ is always false, since $S^{\mathcal{A}} = \emptyset$, so the antecedent of the implication is always false, so for all values of x and y the formula $S(x,y) \to R(f(y))$ is true.

On the other hand, in $\mathcal{B}$, for the formula to be true it needs to be true for the value of 0 for x, for instance. But for 0, we have $\exists y(S(0,y) \to R(f(y)))$. Note that for any value for y the consequent is false, since 0 is not prime. But is $S(0,y)$ always true? Yes. So the formula is always false. For the rest of the formulas we skip the explanation.

b) $\exists z(R(f(z)) \wedge S(z, g(z,z)))$

Solution: $\mathcal{A} = (\mathbb{N}, \cdot^{\mathcal{A}})$ where:
$a^{\mathcal{A}} = 1 \qquad f^{\mathcal{A}}(x) = x \times 0 \qquad g^{\mathcal{A}}(x,y) = x + y$
$R^{\mathcal{A}} = \{x \mid x \geq 1\} \qquad S^{\mathcal{A}} = \{(x,y) \mid x \leq y\}$

$\mathcal{B} = (\mathbb{N}, \cdot^{\mathcal{B}})$ where:
$a^{\mathcal{B}} = 1 \qquad f^{\mathcal{B}}(x) = x \times 0 \qquad g^{\mathcal{B}}(x,y) = x + y$
$R^{\mathcal{B}} = \emptyset \qquad S^{\mathcal{B}} = \emptyset$

c) $R(a) \to S(a,a)$

Solution: $\mathcal{A} = (\mathbb{N}, \cdot^{\mathcal{A}})$ where:
$a^{\mathcal{A}} = 1 \qquad f^{\mathcal{A}}(x) = x \times 0 \qquad g^{\mathcal{A}}(x,y) = x + y$
$R^{\mathcal{A}} = \{x \mid x \geq 1\} \qquad S^{\mathcal{A}} = \{(x,y) \mid x \leq y\}$

$\mathcal{B} = (\mathbb{N}, \cdot^{\mathcal{B}})$ where:
$a^{\mathcal{B}} = 1 \qquad f^{\mathcal{B}}(x) = x \times 0 \qquad g^{\mathcal{B}}(x,y) = x + y$
$R^{\mathcal{B}} = \mathbb{N} \qquad S^{\mathcal{B}} = \emptyset$

d) $\forall x(R(x)) \to \forall y(S(g(y,a), f(y)))$

Solution: $\mathcal{A} = (\mathbb{N}, \cdot^{\mathcal{A}})$ where:
$a^{\mathcal{A}} = 1 \qquad f^{\mathcal{A}}(x) = x \times 0 \qquad g^{\mathcal{A}}(x,y) = x + y$
$R^{\mathcal{A}} = \emptyset \qquad S^{\mathcal{A}} = \{(x,y) \mid x \geq y\}$

$\mathcal{B} = (\mathbb{N}, \cdot^{\mathcal{B}})$ where:
$a^{\mathcal{B}} = 1 \qquad f^{\mathcal{B}}(x) = x \times 0 \qquad g^{\mathcal{B}}(x,y) = x + y$
$R^{\mathcal{B}} = \mathbb{N} \qquad S^{\mathcal{B}} = \{(x,y) \mid x < y\}$

4.3) Show that the following formulas are not semantically equivalent, indicating a language $\mathcal{L}$, formulas φ and ψ of $\mathcal{L}$, and a structure $\mathcal{A} = (A, \cdot^{\mathcal{A}})$ of $\mathcal{L}$, such that $\mathcal{A}$ is not a model of the corresponding formula.

a) $\forall x\,(\varphi \vee \psi)$ and $\forall x\,\varphi \vee \forall x\,\psi$

Solution: Consider a language $\mathcal{L}$ with two unitary predicates $\mathcal{R}_1 = \{P, Q\}$ and let $\varphi = P(x)$ and $\psi = Q(x)$. Now consider a structure $\mathcal{A} = (\{1,2\}, \cdot^{\mathcal{A}}))$ and such that $P^{\mathcal{A}} = \{1\}$ and $Q^{\mathcal{A}} = \{2\}$, and any variable interpretation s. Then for every $a \in \{1,2\}$ it is true that $a \in \{1\}$ or $a \in \{2\}$, which means that for every $a \in \{1,2\}$ $\mathcal{A} \models_{s[x \mapsto a]} (P(x) \vee Q(x))$, that is $\mathcal{A} \models_s \forall x\,(P(x) \vee Q(x))$. On the other hand there exists an $a \in \{1,2\}$ such that $a \notin \{1\}$, which implies there exists an $a \in \{1,2\}$, such that $\mathcal{A} \not\models_{s[x \mapsto a]} P(x)$, therefore $\mathcal{A} \not\models_s \forall x\,P(x)$. And there exists an $b \in \{1,2\}$ such that $b \notin \{2\}$, which implies there exists an $b \in \{1,2\}$, such that $\mathcal{A} \not\models_{s[x \mapsto b]} Q(x)$, therefore $\mathcal{A} \not\models_s \forall x\,Q(x)$. Therefore $\mathcal{A} \not\models_s \forall x\,P(x) \vee \forall x\,Q(x)$

b) $\exists x\,\varphi \wedge \exists x\,\psi$ and $\exists x\,(\varphi \wedge \psi)$

Solution: Consider the same language $\mathcal{L}$ as in (a) with the same unitary predicates $\mathcal{R}_1 = \{P, Q\}$ and $\varphi = P(x)$ and $\psi = Q(x)$, as well as the same structure $\mathcal{A} = (\{1,2\}, \cdot^{\mathcal{A}}))$ and such that $P^{\mathcal{A}} = \{1\}$ and $Q^{\mathcal{A}} = \{2\}$, and any variable interpretation s. There exists an $a \in \{1,2\}$ such that $a \in \{1\}$ therefore $\mathcal{A} \models \exists x\,\varphi$. Similarly, there exists an $a \in \{1,2\}$ such that $a \in \{2\}$ therefore $\mathcal{A} \models \exists x\,\psi$. Therefore $\mathcal{A} \models_s \exists x\,\varphi \wedge \exists x\,\psi$. On the other hand for every $a \in \{1,2\}$ either $a \notin \{1\}$ or $a \notin \{2\}$. Therefore there there is no $a \in \{1,2\}$ such that $a \in \{1\}$ and $a \in \{2\}$, which implies $\mathcal{A} \not\models_s \exists x\,(\varphi \wedge \psi)$.

c) $\forall x\,(\varphi \to \psi)$ and $\exists x\,\varphi \to \forall x\,\psi$

Solution: Consider the same language $\mathcal{L}$ as in (a) with the same unitary predicates $\mathcal{R}_1 = \{P, Q\}$ and $\varphi = P(x)$ and $\psi = Q(x)$. Now consider the structure $\mathcal{A} = (\{1,2\}, \cdot^{\mathcal{A}})$ and such that $P^{\mathcal{A}} = Q^{\mathcal{A}} = \{2\}$, and any variable interpretation s. Then for every $a \in \{1,2\}$ we have $a \notin \{2\}$ or $a \in \{2\}$, that is for every $a \in \{1,2\}$, we gave $\mathcal{A} \not\models_{s[x\mapsto a]} \varphi$ or $\mathcal{A} \models_{s[x\mapsto a]} \psi$, which implies $\mathcal{A} \models_s \forall x\,(\varphi \to \psi)$. On the other hand there exists an $a \in \{1,2\}$ such that $a \in \{2\}$, therefore $\mathcal{A} \models_s \exists x\,\varphi$. And there exists an $a \in \{1,2\}$ such that $a \notin \{2\}$, therefore $\mathcal{A} \not\models_s \forall x\,\psi$. Therefore $\mathcal{A} \not\models_s \exists x\,\varphi \to \forall x\,\psi$.

d) $\exists x\,(\varphi \to \psi)$ and $\exists x\,\varphi \to \exists x\,\psi$

Solution: Consider the same language $\mathcal{L}$ as in (a) with the same unitary predicates $\mathcal{R}_1 = \{P, Q\}$ and $\varphi = P(x)$ and $\psi = Q(x)$. Now consider the structure $\mathcal{A} = (\{1,2\}, \cdot^{\mathcal{A}})$ and such that $P^{\mathcal{A}} = \{1\}$ and $Q^{\mathcal{A}} = \{\}$. Now we can see that $\mathcal{A} \models \exists x(P(x) \to Q(x))$, since we can choose 2 for x and get $\mathcal{A} \models P(2) \to Q(2)$, because $P(2)$ is false. However, $\mathcal{A} \models \exists x P(x)$ and also $\mathcal{A} \not\models \exists x Q(x)$. So we get $\mathcal{A} \not\models \exists P(x) \to \exists x Q(x)$.

e) $\forall x\,(\varphi \to \psi)$ and $\forall x\,\varphi \to \forall x\,\psi$

Solution: Consider the same language $\mathcal{L}$ as in (a) with the same unitary predicates $\mathcal{R}_1 = \{P, Q\}$ and $\varphi = P(x)$ and $\psi = Q(x)$. Now consider the structure $\mathcal{A} = (\{1,2\}, \cdot^{\mathcal{A}})$ and such that $P^{\mathcal{A}} = \{1\}$ and $Q^{\mathcal{A}} = \{2\}$. Now we can see that $\mathcal{A} \models \forall x P(x) \to \forall x Q(x)$ since for the antecedent of the implication $\mathcal{A} \not\models \forall x P(x)$. However, we have that $\mathcal{A} \not\models P(1) \to Q(1)$ since $\mathcal{A} \models P(1)$, but $\mathcal{A} \not\models Q(1)$. Therefore, we have that $\mathcal{A} \not\models \forall x(P(x) \to Q(x))$.

f) $\forall x \exists y\,\varphi$ and $\exists y \forall x\,\varphi$

Solution: Consider the language $\mathcal{L}$ with a binary predicate $\mathcal{R}_2 = \{P\}$ and $\varphi = P(x, y)$. Now consider the structure $\mathcal{A} = (\{1,2\}, \cdot^{\mathcal{A}})$ and such that $P^{\mathcal{A}} = \{(1,1), (2,2)\}$, and any variable interpretation s. Now, for every $a \in \{1,2\}$ there is a $b \in \{1,2\}$ such that $(a,b) \in \{(1,1),(2,2)\}$ (just consider $b = a$). This implies that for every $a \in \{1,2\}$ there is a $b \in \{1,2\}$ such that $\mathcal{A} \models_{s[x\mapsto a][y\mapsto b]} P(x,y)$, therefore $\mathcal{A} \models_s \forall x \exists y\,\varphi$. On the other hand, for every $a \in \{1,2\}$ there is a $b \in \{1,2\}$ such that $(a,b) \notin \{(1,1),(2,2)\}$ (just consider $b \neq a$). This is equivalent to say that for every $a \in \{1,2\}$ there is a $b \in \{1,2\}$ such that $\mathcal{A} \not\models_{s[x\mapsto a][y\mapsto b]} P(x,y)$, which is equivalent to saying that it is not true that there exists an $a \in \{1,2\}$ that for every $b \in \{1,2\}$ is such that $\mathcal{A} \models_{s[x\mapsto a][y\mapsto b]} P(x,y)$, therefore $\mathcal{A} \not\models_s \exists y \forall x\,\varphi$.

4.4) For each of the following formulas determine an equivalent prenex normal form.

(a) $\neg\forall x\,R(x,u) \wedge \exists y\,(\neg\forall x\,R(y,x) \to Q(u,y))$

Solution: First we remove implications using semantic equivalence laws:

$$\begin{aligned} \neg\forall x\,R(x,u) \wedge \exists y\,(\neg\forall x\,R(y,x) \to Q(u,y)) &\Leftrightarrow \neg\forall x\,R(x,u) \wedge \exists y\,(\neg\neg\forall x\,R(y,x) \vee Q(u,y)) \\ &\Leftrightarrow \neg\forall x\,R(x,u) \wedge \exists y\,(\forall x\,R(y,x) \vee Q(u,y)) \end{aligned}$$

Now we rename bound variables so that no variable occurs bound and free and all bound variables are distinct:

$$\neg\forall x\,R(x,u) \wedge \exists y\,(\forall z\,R(y,z) \vee Q(u,y))$$

Now we apply semantic equivalence laws until the formula is in prenex normal form:

$$\begin{aligned} \neg\forall x\,R(x,u) \wedge \exists y\,(\forall z\,R(y,z) \vee Q(u,y)) &\Leftrightarrow \exists x\,\neg R(x,u) \wedge \exists y\,\forall z\,(R(y,z) \vee Q(u,y)) \\ &\Leftrightarrow \exists x\,\exists y\,\forall z\,(\neg R(x,u) \wedge (R(y,z) \vee Q(u,y)) \end{aligned}$$

Note that the following is also a valid prenex normal form for this formula:

$$\exists y\,\forall z\,\exists x\,(\neg R(x,u) \wedge (R(y,z) \vee Q(u,y))$$

(b) $\forall x\, \neg \exists y \forall u\, P(x, y) \rightarrow \neg(\exists z\, P(x, y) \rightarrow \forall y\, P(u, y))$

Solution: First we remove implications using semantic equivalence laws:

$$\begin{aligned} \forall x\, \neg \exists y \forall u\, P(x, y) \rightarrow \neg(\exists z\, P(x, y) \rightarrow \forall y\, P(u, y)) &\Leftrightarrow \neg \forall x\, \neg \exists y \forall u\, P(x, y) \vee \neg(\exists z\, P(x, y) \rightarrow \forall y\, P(u, y)) \\ &\Leftrightarrow \neg \forall x\, \neg \exists y \forall u\, P(x, y) \vee \neg(\neg \exists z\, P(x, y) \vee \forall y\, P(u, y)) \end{aligned}$$

Now we rename bound variables so that no variable occurs bound and free and all bound variables are distinct:

$$\neg \forall w\, \neg \exists r \forall s\, P(w, r) \vee \neg(\neg \exists z\, P(x, y) \vee \forall t\, P(u, t))$$

Now we apply semantic equivalence laws until the formula is in prenex normal form:

$$\begin{aligned} \neg \forall w\, \neg \exists r \forall s\, P(w, r) \vee \neg(\neg \exists z\, P(x, y) \vee \forall t\, P(u, t)) &\Leftrightarrow \exists w\, \exists r\, \forall s\, P(w, r) \vee (\exists z\, P(x, y) \wedge \exists t\, \neg P(u, t)) \\ &\Leftrightarrow \exists w\, \exists r\, \forall s\, \exists z\, \exists t\, (P(w, r) \vee (P(x, y) \wedge \neg P(u, t))) \end{aligned}$$

(c) $\exists x\, S(x, x) \vee \exists z\, (P(z) \wedge \forall x\, R(x, z))$

Solution: In this case we start by renaming bound variables so that no variable occurs bound and free and all bound variables are distinct:

$$\exists x\, S(x, x) \vee \exists z\, (P(z) \wedge \forall y\, R(y, z))$$

Now we apply semantic equivalence laws until the formula is in prenex normal form:

$$\exists x\, S(x, x) \vee \exists z\, (P(z) \wedge \forall y\, R(y, z)) \Leftrightarrow \exists x\, \exists z\, \forall y\, (S(x, x) \vee (P(z) \wedge R(y, z)))$$